

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed Python services to integrate **LLM APIs**, creating automated summarization and classification utilities for technical documentation content.
- Designed and implemented data transformations and metadata models to prepare **DITA XML content** for AI consumption, enabling reliable chunking and indexing for downstream AI systems.
- Built and extended **REST APIs** to power AI-driven documentation experiences, such as search assistants and Q&A; bots, without disrupting existing writer workflows.
- Collaborated with documentation leaders and platform engineers to define requirements for **AI-ready documentation**, focusing on structure, semantics, and metadata enrichment.
- Integrated AI assistance tools into the existing **Oxygen/DITA authoring and publishing workflows** using **Git-based version control** and Azure DevOps pipelines.
- Contributed technical insights to the Documentation Community of Practice regarding best practices for preparing content for AI systems, including schema design and content quality checks.
- Containerized AI tooling services using **Docker** and deployed them to an **AWS EC2** environment, ensuring consistent execution across development and production.

SKILLS

Languages – Python, Java, SQL, R, TypeScript, JavaScript, Shell Scripting

AI/ML & LLMs – LLM APIs, RAG pipelines, AI-assisted workflows, Classification, Tagging, Summarization, Prompt Engineering

Content & Docs Tooling – DITA XML, Structured Content, Oxygen XML Editor, OpenDITA, Git, Azure DevOps Pipelines, Confluence

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, CI/CD

Data & Databases – PostgreSQL, MySQL, Oracle, pandas, NumPy, Data Warehousing, ETL

PROJECTS

AI-Assisted Content Classification Service (HCL Technologies)

- Developed a Python service to automatically classify and tag incoming technical documentation drafts based on content similarity and predefined categories, improving content organization efficiency by approximately 30%.
- Integrated the classification service with a **Git repository**, triggering classification upon new commit events and storing metadata alongside documentation files.
- Engineered the service to accept and process **DITA XML** documents, extracting relevant text content for analysis by machine learning models.

DITA XML to AI-Ready Data Pipeline (HCL Technologies)

- Designed and implemented a pipeline using Python to transform structured **DITA XML** documentation into a format suitable for AI indexing and retrieval, focusing on semantic markup and metadata extraction.
- Developed scripts to parse DITA maps and topics, generating structured JSON output that preserves hierarchical relationships and key attributes for downstream AI processing.
- Applied data quality checks to ensure consistency and completeness of the transformed data, reducing data ingestion errors for AI models by an estimated 15%.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)